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Asymptotic Analysis of Searching Algorithms: Performance Evaluation of Linear Search and Binary Search Using Big-O, Big-Theta, and Big-Omega Notations

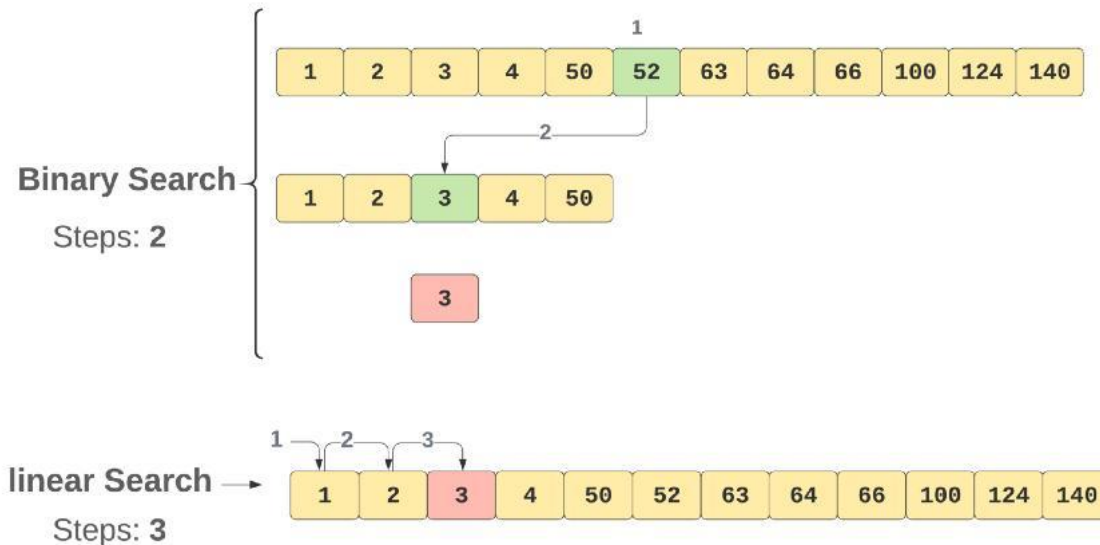
CSA0620- DESIGN AND ANALYSIS OF ALGORITHM

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INTRODUCTION

- Searching algorithms are used to find a specific element in a dataset. **Linear Search** checks each element one by one, while **Binary Search** repeatedly divides a **sorted** dataset into halves to locate the target more efficiently. Asymptotic analysis using **Big-O**, **Big-Theta**, and **Big-Omega** notations helps compare their performance and identify the most efficient algorithm for



Problem Statement

- Efficient searching is essential for applications that process large amounts of data. The challenge is to determine which searching algorithm provides better performance as the dataset size increases. This presentation analyzes the efficiency of **Linear Search** and **Binary Search** using asymptotic notation and compares their execution times under best, average, and worst-case scenarios.

ALGORITHM OVERVIEW

- **Linear Search**
 - Searches elements one by one from the beginning.
 - Stops when the target is found or the list ends.
 - Works with both sorted and unsorted datasets.
- **Binary Search**
 - Works only on sorted datasets.
 - Compares the target with the middle element.
 - Repeatedly halves the search space until the element is found.

Algorithm / Pseudocode

• Algorithm LinearSearch(A, n, key)

1. for $i = 0$ to $n-1$
2. if $A[i] == \text{key}$
3. return i
4. return -1

Algorithm BinarySearch(A, low, high, key)

1. while $\text{low} \leq \text{high}$
2. $\text{mid} = (\text{low} + \text{high})/2$
3. if $A[\text{mid}] == \text{key}$
4. return mid
5. else if $\text{key} < A[\text{mid}]$
6. $\text{high} = \text{mid} - 1$
7. else
8. $\text{low} = \text{mid} + 1$
9. return -1

Linear Search Algorithm

Element to search for: 4

Iteration 1: Begin search from index 0 and check $\text{arr}[0] == \text{element}$

arr[] =	1	2	4	6	8
Array Indices →	0	1	2	3	4

Iteration 2: Move to next element and check $\text{arr}[1] == \text{element}$

	1	2	4	6	8
Array Indices →	0	1	2	3	4

Iteration 3: Move to next element and check $\text{arr}[2] == \text{element}$

	1	2	4	6	8
Array Indices →	0	1	2	3	4

Element Found!

Binary Search

	0	1	2	3	4	5	6	7	8	9
Search 23	2	5	8	12	16	23	38	56	72	91
	L=0				M=4					H=9
23 > 16 take 2 nd half	2	5	8	12	16	23	38	56	72	91
						L=5		M=7		H=9
23 < 56 take 1 st half	2	5	8	12	16	23	38	56	72	91
						L=5		M=6		H=9
Found 23, Return 5	2	5	8	12	16	23	38	56	72	91



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WORKING EXAMPLE (DRY RUN)



- **Dataset**
- **[10, 20, 30, 40, 50, 60, 70]**
- **Target = 50**
- **Linear Search**
- **10 → 20 → 30 → 40 → 50 ✓**
- **Comparisons = 5**
- **Binary Search**
- **Middle = 40**
- **50 > 40**
- **Search Right Half**
- **Middle = 60**
- **50 < 60**
- **Search Left Half**
- **Middle = 50 ✓**
- **Comparisons = 3**

COMPLEXITY ANALYSIS

Algorithm	Best Case	Average Case	Worst Case
Linear Search	$\Omega(1)$	$\Theta(n)$	$O(n)$
Binary Search	$\Omega(1)$	$\Theta(\log n)$	$O(\log n)$

OBSERVATION:

- **Big-O** represents the worst-case performance.
- **Big-Theta** represents the average-case performance.
- **Big-Omega** represents the best-case performance.
- Binary Search performs significantly better for large datasets.



COMPARATIVE ANALYSIS



Feature	Linear Search	Binary Search	Feature
Dataset Requirement	Sorted or Unsorted	Sorted Only	Dataset Requirement
Best Case	$O(1)$	$O(1)$	Best Case
Average Case	$O(n)$	$O(\log n)$	Average Case
Worst Case	$O(n)$	$O(\log n)$	Worst Case

Linear Search

$100 \rightarrow 99 \rightarrow 98 \rightarrow \dots \rightarrow 1$

Search Space Reduction

Binary Search

100

↓

50

↓

25

↓

12

↓

6

↓

3

↓

1

APPLICATIONS

- **Linear Search**
- Small datasets
- Unsorted lists
- Simple database searches
- Finding elements in arrays
- **Binary Search**
- Search engines
- Database indexing
- Dictionary applications
- Library management systems
- File systems

RESULTS & DISCUSSION

Results

- Linear Search requires up to **n comparisons**.
- Binary Search requires only **$\log_2 n$ comparisons**.
- Binary Search reduces the search space by half in every iteration.
- Performance improvement becomes significant as dataset size increases.

Discussion

- Experimental analysis shows that **Linear Search** is suitable for small or unsorted datasets because of its simplicity. However, **Binary Search** provides much better performance on large sorted datasets due to its logarithmic time complexity. Therefore, Binary Search is preferred when the data is already sorted.

CONCLUSION

- Asymptotic analysis demonstrates that **Binary Search** is more efficient than **Linear Search** for large datasets. Linear Search has **$O(n)$** worst-case complexity, whereas Binary Search achieves **$O(\log n)$** complexity by repeatedly reducing the search space. Therefore, Binary Search is the preferred algorithm for searching in sorted datasets, while Linear Search remains useful for small or unsorted collections.

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THANK YOU..



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